



SOFTWARE REQUIREMENT SPECIFICATION

Quiz Application System

(Group 3)

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I. Record of Changes

Date	A* M, D	In charge	Change Description

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Overall Requirements

1.1 Context Diagram

Quiz Application System is a software system that enables users to create, manage, and take online multiple-choice quizzes. The system supports two main business functions:

- Quiz – taking: Registered users (User) can browse the list of available quizzes, complete a quiz within a set time limit, and receive automatically calculated results immediately after submission.
- Quiz – management: Administrators (Admin) are responsible for creating, updating, and removing quizzes, questions, and answers to ensure the question bank remains accurate and up to date.

In addition, the system serves unregistered visitors (Guest) at the initial access stage, allowing them to preview the list of public quizzes before deciding to register an account. The system operates as a standalone application with internal data storage and does not yet integrate with external systems such as payment gateways or Learning Management Systems (LMS). The Context Diagram below defines the system boundary and all external entities that interact with it.

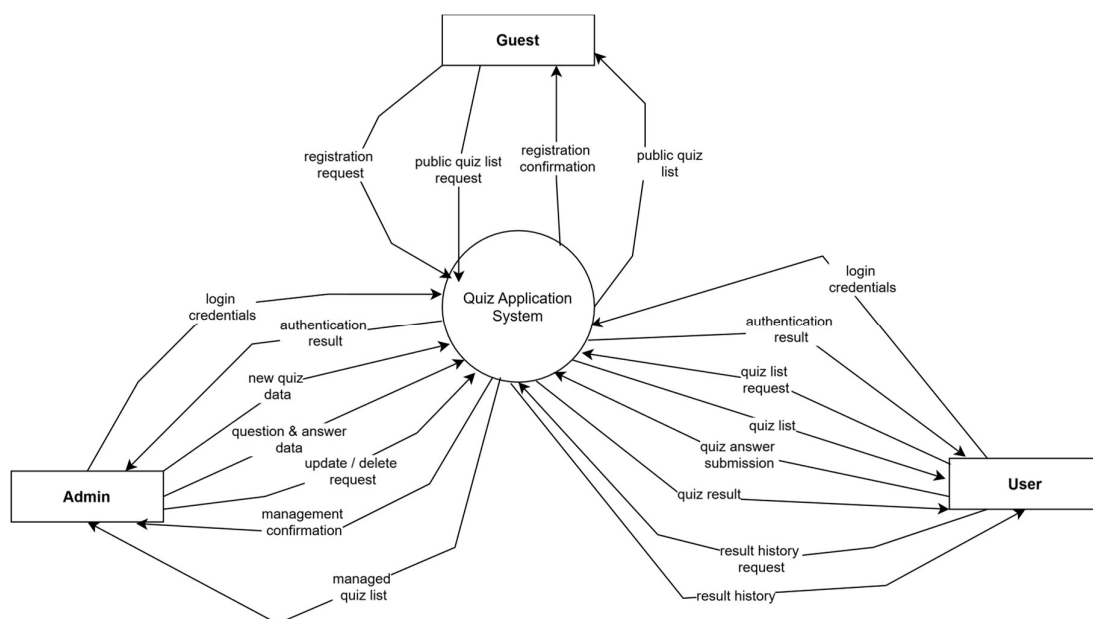


Figure 1: Context Diagram of Quiz Application System

1.2 Main Business Processes

This section describes the core business workflows of the Quiz Application System, presented as swim-lane diagrams to clarify the responsibilities and hand-offs between actors and the system.

1.2.1 Quiz Taking Process

This process describes how a User selects and completes a quiz. After browsing the quiz list, the User selects a quiz, and the system verifies whether it is still active. If active, the system displays the questions and starts the timer; the User answers and submits the quiz, after which the system automatically calculates the score and displays the result. If the quiz is no longer active, the system shows an error and the process ends.

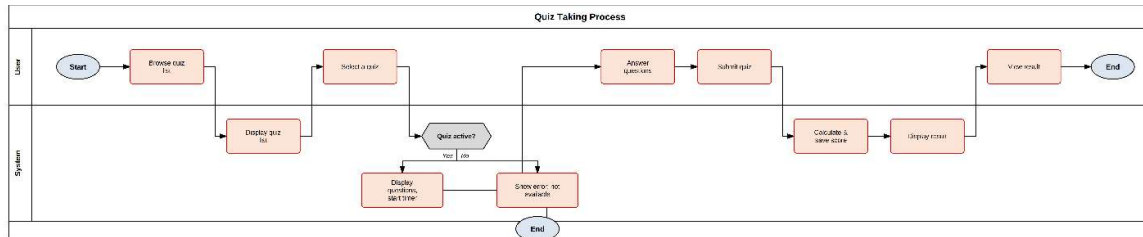


Figure 2: Quiz Taking Process Diagram

1.2.2 Quiz Management Process

This process describes how an Admin creates and publishes a quiz. After logging in, the Admin enters the quiz information and then chooses how to add questions: either by selecting existing questions from the shared question bank, or by creating new questions with their answers. The system validates the data; if valid, the quiz is saved, its questions are linked, and a confirmation is returned, allowing the Admin to publish it. If the data is invalid, the system returns a validation error and the process ends without saving.

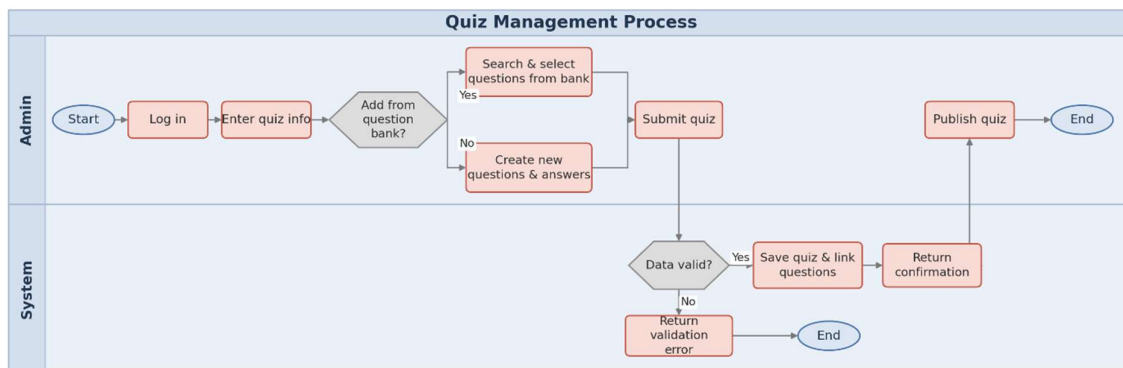


Figure 3: Quiz Management Process Diagram

1.2.3 Guest Onboarding Process

This process describes how a Guest becomes a registered User. A Guest accesses the homepage and views the public quiz list, then enters registration information to create an account. The system validates the data; if valid, the account is created, a confirmation is sent, and the Guest can log in as a User. If the data is invalid, the system returns a validation error and the process ends without creating an account.

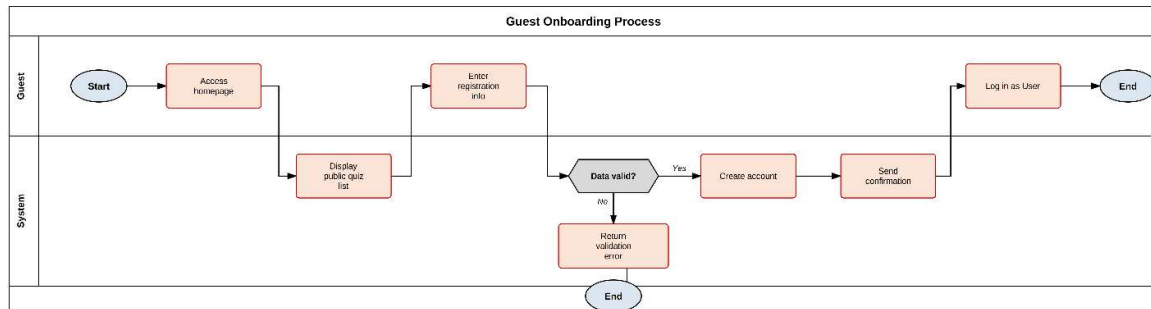


Figure 4: Guest Onboarding Process Diagram

1.3 User Requirements

1.3.1 Actors

#	Actor	Description
1	Guest	An unauthenticated visitor who accesses the Quiz Application System without logging in. A Guest can browse the public quiz list and view general quiz information but cannot take a quiz or access any personalized features. A Guest becomes a User after successfully registering and logging into the system.
2	User	An authenticated individual who has registered an account and logged into the system to take quizzes. A User can view the list of available quizzes, take a quiz within the allotted duration, submit answers, and view their own quiz results. A User has no permission to create, modify, or delete quiz content.
3	Admin	An authenticated staff member with full administrative rights over quiz content. An Admin can create, update, and delete quizzes, as well as manage the questions and answers associated with each quiz. The Admin is responsible for maintaining the accuracy, completeness, and availability of quiz content published to Users.

1.3.2 Use Cases (UC)

ID	Use Case	Feature	Use Case Description
UC01	Register Account	Guest Onboarding	Allows a Guest to create a new account by providing registration information (e.g., username, email, password) so that they can log in and access personalized features as a User.
UC02	View Public Quiz List	Guest Onboarding	Allows a Guest to browse the list of publicly available quizzes without logging in, so they can preview what the system offers before registering.
UC03	Login System	Guest Onboarding	Allows a registered User or Admin to log into the system using valid credentials in order to access authenticated features corresponding to their role.

UC04	View Quiz List	Quiz Taking	Allows a logged-in User to view the list of quizzes available for them to take, including quiz title, description, and duration.
UC05	Take a Quiz	Quiz Taking	Allows a logged-in User to select a quiz, answer its questions within the given duration, and submit their answers for scoring.
UC06	View Quiz Result	Quiz Taking	Allows a logged-in User to view the result of a quiz they have completed, including the score and correct/incorrect answers.
UC07	Create Quiz	Quiz Management	Allows an Admin to create a new quiz by entering quiz information (title, description, duration) together with its questions and answers.
UC08	Update Quiz	Quiz Management	Allows an Admin to modify the information of an existing quiz, such as its title, description, duration, or associated questions/answers.
UC09	Delete Quiz	Quiz Management	Allows an Admin to remove an existing quiz from the system, so it is no longer available to Users.
UC10	Manage Questions & Answers	Quiz Management	Allows an Admin to create, update, or delete questions and their answers in the shared question bank, independently of any specific quiz.
UC11	Log Out	Account Management	Allows a logged-in User or Admin to end their current session and return to the unauthenticated state.
UC12	View/Edit User Profile	Account Management	Allows a logged-in User to view and update their personal profile information.
UC13	Change Password	Account Management	Allows a logged-in User to change their current password from within their profile. (extends UC12)
UC14	Reset Password	Account Management	Allows a User who has forgotten their password to reset it via the Login screen. (extends UC03)
UC15	View Quiz Details	Quiz Taking	Allows a logged-in User to view detailed information of a specific quiz (e.g., number of questions, duration, instructions) before starting it. (extends UC04)
UC16	Add Questions from Question Bank	Quiz Management	Allows an Admin to search the question bank and select existing questions to add into a quiz, instead of creating them from scratch.
UC17	Remove Question from Quiz	Quiz Management	Allows an Admin to remove a question from a quiz without deleting it from the question bank, so other quizzes using it are unaffected.

1.3.3 Use Case Diagrams

1.3.3.1 UCs for Guest

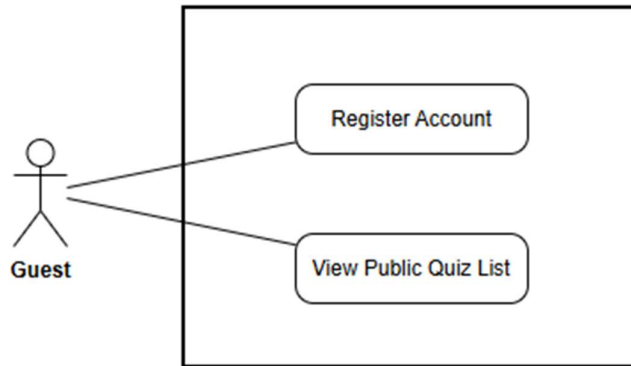


Figure 5: UCs for Guest

1.3.3.2 UCs for User

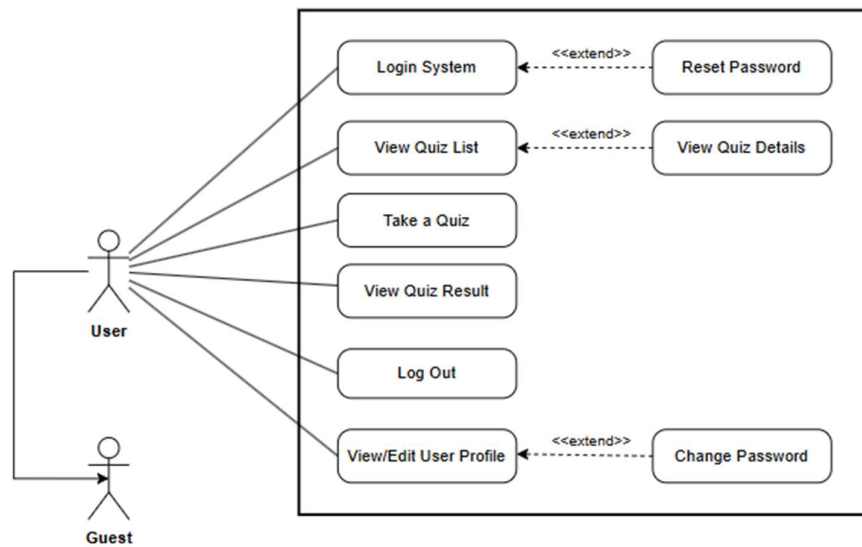


Figure 6: UCs for User

1.3.3.3 UCs for Admin

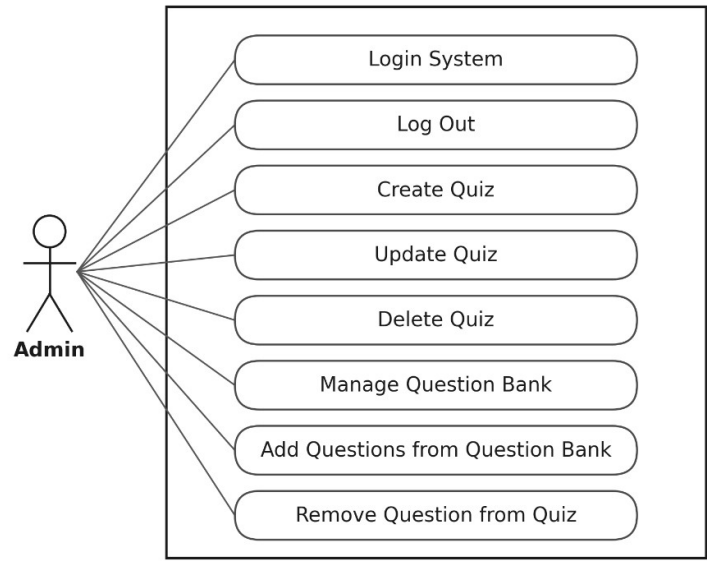


Figure 7: UCs for Admin

1.4 System Functionalities

1.5 Entity Relationship Diagram

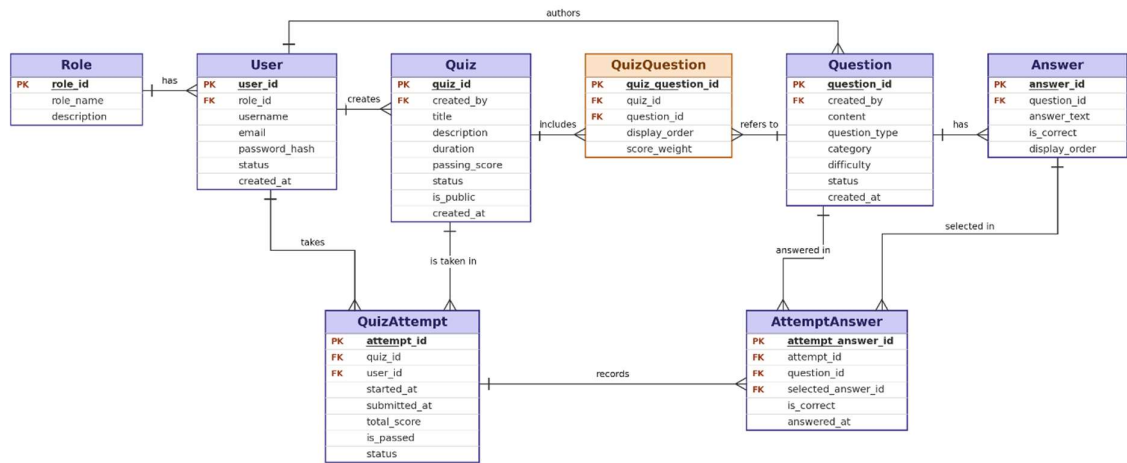


Figure 8: Entity Relationship Diagram

Entities Description

#	Entity	Description
1	Role	Defines the system roles (Admin, User) and the permissions associated with each role.

2	User	Stores registered account information, including credentials, status, and the assigned role.
3	Quiz	Stores quiz information such as title, description, duration, passing score, and publication status.
4	Question	Stores the questions belonging to a quiz, including content, question type, and display order.
5	QuizQuestion	Junction entity linking a quiz to the questions it includes. It also stores quiz-specific settings for each question, such as its display order and score weight within that quiz.
6	Answer	Stores the answer options of each question and indicates which option is correct.
7	QuizAttempt	Records each attempt a User makes on a quiz, including start/submit time, total score, and pass/fail result.
8	AttemptAnswer	Records the answer selected by the User for each question within a quiz attempt, used to generate the result breakdown.

2. Use Case Specifications

2.1 Guest Onboarding

2.1.1 Register Account

Primary Actors	Guest	Secondary Actors	None
Description	As a Guest, I want to create an account so that I can log in and access quiz-taking features as a User.		
Preconditions	The Guest has accessed the Registration screen.		
Postconditions	A new User account is created and stored in the system; a confirmation is returned to the Guest.		
Normal Sequence/Flow	1. Guest clicks Register from the Home Page. 2. System displays the Registration screen. 3. Guest enters registration information (username, email, password, confirm password). 4. Guest clicks the Register button. 5. System validates the input data. 6. System creates the new account and saves it. 7. System returns a confirmation message and redirects the Guest to the Login screen.		
Alternative Sequences/Flows	Step 5_System cannot validate the input Guest cannot register and receives a relevant error message in one of the cases below: 1. Required field(s) left blank. 2. Email is already registered. 3. Password and Confirm Password do not match. 4. Password does not meet the minimum complexity requirement.		

2.1.2 Login System

Primary Actors	User, Admin	Secondary Actors	None
Description	As a User or Admin, I want to log into the system using my credentials so that I can access features according to my role.		
Preconditions	The actor already has a registered and active account.		
Postconditions	The actor is authenticated and redirected to the appropriate screen based on their role (User → Quiz List; Admin → Quiz Management Dashboard).		
Normal Sequence/Flow	1. Actor clicks Login from the Home Page, or attempts to access an authenticated feature. 2. System displays the Login screen. 3. Actor enters email and password. 4. Actor clicks the Login button. 5. System validates the login details. 6. System authenticates the actor and directs them to the corresponding screen based on role.		
Alternative Sequences/Flows	Step 5_System cannot authenticate the actor Actor cannot log in and receives a relevant error message in one of the cases below: 1. Email and/or Password field left blank. 2. Email or Password is incorrect. 3. If the actor enters incorrect credentials 5 times consecutively, the system locks the account for 30 minutes.		

2.2 Quiz Taking

2.2.1 Take a Quiz

Primary Actors	User	Secondary Actors	None
Description	As a User, I want to take a quiz so that I can test my knowledge and receive a score.		
Preconditions	The User has logged in and selected a quiz from the Quiz List.		
Postconditions	The User's answers are recorded, scored, and the result is displayed and saved.		
Normal Sequence/Flow	1. User selects a quiz from the Quiz List. 2. System checks whether the selected quiz is still active. 3. System displays the quiz questions and starts the countdown timer based on the quiz's Duration. 4. User answers the questions. 5. User clicks Submit to finish the quiz. 6. System calculates the score based on the User's answers. 7. System saves the result and displays it to the User.		
Alternative Sequences/Flows	Step 2_Quiz is no longer active 1. System displays an error message informing the User that the quiz is unavailable. 2. System redirects the User back to the Quiz List. The use case ends.		

	Step 3_Time expires before submission 1. System automatically submits the User's current answers when the timer reaches zero. 2. System displays a notice that the time has expired. 3. Return to step 6 of the normal flow.
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2.2.2 View Quiz Result

Primary Actors	User	Secondary Actors	None
Description	As a User, I want to view the result of a quiz I have completed so that I can see my score and review which answers were correct or incorrect.		
Preconditions	The User has completed and submitted a quiz (per UC05 Take a Quiz).		
Postconditions	The quiz result, including score and per-question breakdown, is displayed to the User.		
Normal Sequence/Flow	1. System automatically displays the Quiz Result screen immediately after the User submits the quiz (continuing from UC05), or the User selects a previously completed quiz from their Quiz History. 2. System retrieves the stored result for the selected quiz attempt. 3. System displays the total score, the pass/fail status, and a breakdown of each question showing the User's selected answer versus the correct answer. 4. User may click on a specific question to view a more detailed explanation, if available.		
Alternative Sequences/Flows	Step 2_No result found 1. If the requested quiz attempt does not exist or does not belong to the User, the system displays an error message and redirects the User back to the Quiz List. The use case ends. Step 3_Quiz does not allow reviewing correct answers 1. If the quiz is configured by the Admin to hide correct answers after submission, the system displays only the total score and pass/fail status, without the per-question answer breakdown.		

2.3 Quiz Management

2.3.1 Create Quiz

Primary Actors	Admin	Secondary Actors	None
Description	As an Admin, I want to create a new quiz with its questions and answers so that Users can take it.		
Preconditions	The Admin has logged into the system.		
Postconditions	A new quiz, along with its questions and answers, is saved and made available for publishing.		
Normal Sequence/Flow	1. Admin clicks Create Quiz from the Quiz Management screen. 2. System displays the Quiz Details form. 3. Admin enters the quiz information (title, description, duration).		

	4. Admin adds questions to the quiz using either of two options: - Create a new question with its answers, which is also saved into the question bank; or - Add one or more existing questions selected from the question bank (see UC16). 5. Admin clicks Save. 6. System validates the entered data. 7. System saves the quiz and returns a confirmation. 8. Admin may choose to publish the quiz, making it visible to Users.
Alternative Sequences/Flows	Step 6_System cannot validate the data Admin cannot save the quiz and receives a relevant error message in one of the cases below: - Required field(s) left blank - A question has fewer than two answers, or no answer marked as correct The quiz is not saved and the process ends without changes.

2.3.2 Manage Question Bank

Primary Actors	Admin	Secondary Actors	None
Description	As an Admin, I want to add, update, or delete questions and answers within a quiz so that the quiz content stays accurate and complete.		
Preconditions	The Admin has logged in and opened an existing quiz for editing.		
Postconditions	The quiz's questions and/or answers are updated according to the Admin's changes, while the quiz still satisfies the minimum content requirement.		
Normal Sequence/Flow	- Admin opens the Manage Questions & Answers screen of an existing quiz. - Admin adds, edits, or removes a question or an answer. - Admin clicks Save. - System validates that each question still has at least two answers with one marked correct, and the quiz still has at least one question. - System saves the changes and returns a confirmation.		
Alternative Sequences/Flows	Step 4_System cannot validate the data - If the change would leave a question with fewer than two answers, or leave the quiz with no questions, the system rejects the change and displays an error. - The previous state of the question/answer is retained and the process ends without saving. Step 2_Question is used by multiple quizzes - Before saving an edit or deletion, the system checks how many quizzes currently use this question. - If the question is used by more than one quiz, the system displays a warning listing the affected quizzes and asks the Admin to confirm. - If the Admin cancels, no change is saved and the use case ends. - For deletion, if the question has already been answered in a submitted quiz attempt, the system does not delete it but marks its status as Archived instead, preserving historical results.		

2.3.3 Add Questions from Question Bank

Primary Actors	Admin	Secondary Actors	None
Description	As an Admin, I want to add existing questions from the question bank into a quiz so that I can reuse previously created content instead of retyping it.		
Preconditions	The Admin has logged in and opened a quiz for editing. At least one question exists in the question bank.		
Postconditions	The selected questions are linked to the quiz with their display order and score weight; the questions themselves remain unchanged in the bank.		
Normal Sequence/Flow	1. Admin clicks Add from Question Bank on the Quiz Details screen. 2. System displays the Question Bank screen with search and filter options (by keyword, category, difficulty, question type). 3. Admin filters or searches for questions and previews their content and answers. 4. Admin selects one or more questions and clicks Add to Quiz. 5. System validates the selection (BR-11). 6. System creates a link between the quiz and each selected question, assigning a default display order and score weight. 7. System returns a confirmation and displays the updated question list of the quiz. 8. Admin may adjust the display order and score weight of each added question.		
Alternative Sequences/Flows	Step 5_ Question already exists in the quiz 1. System skips any question already linked to this quiz and notifies the Admin which ones were skipped (BR-11). Remaining questions are added normally. Step 3_ No matching question found 1. System displays an empty-result message and suggests that the Admin create a new question instead. The use case ends.		